

Predicting Day-to-Day Maximum-Temperature Differences at One Station

Evidence from 18 European Stations

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Abstract

Next-day temperature forecasts on dense station networks can look excellent for the wrong reason. Day-to-day persistence and the annual cycle together account for over 90% of the variance in raw temperature series, so a model inherits their skill without learning anything about the atmosphere. This report recasts the problem for a public European multi-station dataset (18 stations, 2000–2010, 163 features) as forecasting the *day-to-day change* in daily maximum temperature at Basel, $\Delta T_{\max}(t+1) = T_{\max}(t+1) - T_{\max}(t)$. In this target space the persistence forecast is identically zero, so skill against the zero-change baseline measures meteorological information rather than calendar knowledge. A ladder of models is fitted under a strictly chronological split (train 2000–2007, validation 2008, test 2009) with training-only preprocessing statistics and moving-block bootstrap uncertainty. All learned models clear the zero-change bar (test MAE 2.393°C) significantly: ridge 1.827°C, Lasso 1.773°C, and an MLP ensemble 1.751°C, with the differences among them inside bootstrap noise. Ablations locate the usable signal outside the target station: its own temperature observations give +0.0% skill. The signal is spread across the field rather than concentrated in a few channels, and the fitted weight mass falls on pressure features. Pooling all 18 stations into one model with no station conditioning underperforms 18 independent models by 0.36°C.

1 Introduction

Public weather datasets invite an apparently easy benchmark: predict tomorrow’s temperature from today’s observations across many stations. On the dataset used here, daily observations from 18 European locations over eleven years, the rule “tomorrow equals today” reaches a test MAE of 1.667°C with $R^2 = 0.9206$ on the raw daily mean temperature. The monthly climatology alone accounts for 74.4% of that series’ variance. Any regression model, however trivial, therefore posts high accuracy on the raw target, and the number certifies the calendar and the persistence of the atmosphere rather than the model.

What remains once that skill is taken away is the question worth asking. We reformulate the task so the trivial forecasts carry no information by construction, and take as target the *day-to-day change* in daily maximum temperature at Basel, $\Delta T_{\max}(t+1)$. The persistence forecast in this space is identically zero, so accuracy beyond that bar has to come from the atmosphere.

The usable signal lies upstream. Differencing already removes the seasonal cycle, and against Basel’s next-day change the strongest same-day predictors are observations elsewhere: on the mean-temperature change Maastricht, Tours and De Bilt correlate 0.35–0.40, and against the maximum-temperature change used here Tours leads at 0.35, followed by Maastricht at 0.20 and De Bilt at 0.19. Sections 3–5 measure how much of that signal survives into a fitted model, and in what form.

2 Dataset

We use the weather prediction dataset of Huber et al. [2] (MIT licence), derived from the European Climate Assessment [1], at release commit 83d70ee. The main table covers 3654 consecutive days from 2000-01-01 to 2010-01-01 across 18 European stations, holding `DATE`, `MONTH` and 163 features. These are 18 stations $\times$ 11 variables (temperatures, humidity, pressure, wind, radiation, sunshine, cloud cover, precipitation), of which 198 station-variable cells are theoretically possible and 35 are absent entirely. `wind_gust` reaches only 7 stations, and Basel reports neither wind speed nor gusts.

Processing applied upstream. The release is clean only by construction. Columns missing more than 5% of values were dropped outright; columns missing at most 5% were mean-imputed by the dataset authors. Both operations are documented upstream and both matter here: mean imputation compresses variance and inflates correlations in the affected columns, while dropping sparse columns discards real if patchy information such as wind gusts. Units are heterogeneous by design. Pressure is expressed in units of 1000 hPa with a standard deviation near 0.008 in the level columns and 0.005 in their changes, humidity runs from 0 to 1, and wind gusts reach 41 m/s, so standardisation is not optional for distance- or gradient-based models.

Audit of physical plausibility. Because upstream imputation fills defects instead of flagging them, every cell was checked against physical ranges. Seventy-seven values cannot occur: three `STOCKHOLM_cloud_cover` cells (two sentinels at -99 , one at 9 oktas against a 0–8 range), 29 days of negative sunshine at Stockholm, four pressures at -0.099 or 0.0003 , and 41 days on which `temp_min` exceeds `temp_max` (40 at Heathrow, one at Roma). They account for 0.013% of 595,602 cells, and the damage concentrates at two stations rather than spreading evenly. One defective cell, Stockholm cloud cover on 2009-06-25, falls inside the test period, so leaving it in place would place a corrupted day in the evaluation set. Differencing amplifies such values into artefacts of order 100 in the affected units, which is why they are removed before any difference is taken.

Structure. The monthly climatology of Basel’s daily mean temperature explains 74.4% of its variance, and the deseasonalised anomaly retains a lag-1 autocorrelation of 0.819, but the change series is close to white: its lag-1 autocorrelation is $+0.07$ for T_{mean} and -0.08 for T_{max} . The feature matrix is heavily redundant, with 62 of 163 features belonging to pairs correlated above 0.95 in absolute value, the strongest being 0.996 between the pressures of two neighbouring stations. Cross-station information is the remaining candidate source of skill, since the target station’s own present-day change correlates with its next-day change at only -0.08 .

3 Problem Formulation

3.1 Three candidate targets and their difficulty floors

For a temperature series the prediction problem changes qualitatively with the object predicted. Table 1 gives each candidate its difficulty floor and what a strong score on it would certify.

Table 1: Three candidate targets for next-day temperature prediction at Basel, and what a strong score on each would certify.

Target	Seasonal share	Persistence score	High score certifies
Raw $T_{\text{mean}}(t+1)$	74.4% of variance	MAE 1.667°C ($R^2 = 0.9206$)	calendar + inertia
Deseasonalised anomaly	removed by design	RMSE 2.288°C	anomaly persistence
$\Delta T_{\text{max}}(t+1)$	≈ 0 by construction	MAE 2.393°C (zero-change)	meteorology, if anything

Predicting the raw value draws on two sources of skill at once. Predicting the anomaly removes the calendar but leaves anomaly persistence, which a persistence forecast already captures: persistence scores 2.288°C RMSE against 3.378°C for the climatological-anomaly null. Those two figures come from a smaller exploratory fit on uncleaned data, with a scikit-learn ridge and a T_{mean} anomaly target. They indicate the size of the effect rather than an MAE that can be set beside the figures in Table 2. Predicting the change removes both sources, and pays for it with a much lower ceiling, since a near-white target cannot support high scores.

3.2 Formal definition

Let $x_t \in \mathbb{R}^{163}$ collect all raw observations on day t across the 18 stations, together with the engineered blocks of Section 4.2. The target is

$$y_t = \Delta T_{\text{max}}^{\text{BASEL}}(t+1) = T_{\text{max}}^{\text{BASEL}}(t+1) - T_{\text{max}}^{\text{BASEL}}(t), \quad (1)$$

and we learn $f : \mathbb{R}^p \rightarrow \mathbb{R}$ by minimising an empirical risk

$$\frac{1}{n} \sum_t \ell(f(x_t), y_t)$$

, where ℓ is the squared error for the linear models and the Huber loss for the MLP, plus an optional penalty. T_{max} is used because it is available at all 18 stations, unlike `temp_min`, which Budapest does not report.

3.3 Equivalence argument: moving persistence into the evaluation

Changing the target does not change the hypothesis space. When $T_{\text{max}}(t)$ is among the inputs, any predictor of tomorrow’s level becomes a predictor of the change by subtracting that input, and the converse holds, so the two tasks share the same hypothesis space. What changes is the evaluation.

In level space the strongest baseline is persistence, $\hat{T}(t+1) = T(t)$, and every model is judged against it. In change space that same baseline is the constant zero, so persistence skill is removed from every model by construction. The identity is exact: a model that copies $T_{\text{max}}(t)$ scores MAE 2.393°C in level space, and the zero forecast scores exactly 2.393°C in change space, because the change is the difference of the two levels and the error is the same quantity either way. What the reformulation changes is that the baseline becomes part of the score rather than a separate comparison, so a model cannot look accurate by inheriting the calendar and the atmosphere’s inertia.

3.4 Alternative tasks

Six alternative targets were set aside. Next-day raw temperature is persistence-contaminated and is already covered by the dataset’s own teaching notebooks and public kernels. Sunshine-duration regression is those notebooks’ headline experiment. Picnic suitability and month classification are derived labels: the picnic flag is a deterministic rule (“no rain and warm”), constant false at one station, with a monthly positive rate spanning two orders of magnitude, so a randomised split degenerates the task into guessing the calendar. Precipitation amount is 35–75% zero-inflated per station and would need a hurdle treatment. Next-day precipitation occurrence is a reasonable low-skill extension, left aside for scope.

3.5 Evaluation protocol

Splitting. Raw daily temperature has a lag-1 autocorrelation of 0.957, so any randomised split leaks. We use one fixed chronological split: train 2000–2007 (2922 days, 2920 samples after

alignment), validation 2008 (366), test 2009-01-01 through 2010-01-01 (366). The final day is a single boundary observation, so the window is effectively calendar 2009 plus one day. Masks are defined on the target date, so a training sample’s label falls inside the training period. The validation year decides every model and hyperparameter choice.

Statistics are training-only. Cleaning thresholds, imputation medians, standardisation parameters, monthly climatologies and regularisation strengths are all estimated on 2000–2007 and applied unchanged to 2008 and 2009.

Metrics. MAE in °C is primary, because the target is heavy-tailed: frontal passages produce $|\Delta T_{\max}| > 10^\circ\text{C}$, and the largest single change in the test period is 11.9°C . We also report RMSE, R^2 in change space, the Pearson correlation between predicted and observed changes (undefined for near-constant forecasts, shown as “—”), and

$$\text{skill} = 1 - \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{zero}}}, \quad (2)$$

the fractional MAE reduction against the zero-change baseline.

Uncertainty. At 366 test days the standard error of MAE is roughly 0.11°C under independent sampling and 0.15°C under the 7-day block bootstrap actually used here, so differences of a few tenths of a degree are marginal. All intervals are moving-block bootstrap 95% CIs with 7-day blocks, chosen because synoptic features persist for several days, 1000 resamples and a fixed seed. Differences between two models use paired bootstraps on the same resampled blocks, which roughly halves the noise on a difference compared with comparing two marginal intervals. Test metrics are computed for the finally selected configuration of each model family; the MLP grid’s test column is reported for all twelve configurations for transparency and did not enter selection.

4 Model

4.1 Pipeline

One pipeline serves every model. Physically impossible cells are masked to NaN before anything is differenced, so that sentinel values cannot become order-100 spikes in the delta features. Masked and missing cells are then imputed with per-station, per-calendar-month medians estimated on the training years. Features are constructed from the cleaned table, standardised with training-set mean and standard deviation, and passed to the model ladder.

The audit in Section 2 counts 77 defective values, but masking moves more cells than that: each of the 41 days on which `temp_min` exceeds `temp_max` nulls both members, so 118 cells are set to NaN in total. Both members go because retaining one of a contradictory pair is worse than discarding both.

4.2 Feature blocks

- **levels** (163): all day- t observations, standardised.
- **deltas** (163): the same columns differenced over one day, $x_t - x_{t-1}$. These carry the weather-system signal.
- **grads** (7): pressure at each of seven stations upstream of Basel, minus pressure at Basel. The seven are Maastricht, Tours, Düsseldorf, München, Kassel, De Bilt and Heathrow, fixed from a lead-correlation probe before any modelling. That probe ranked stations by the day-ahead correlation of their *level* anomaly with Basel’s, in which they score 0.69–0.80;

six were taken by rank and Heathrow by westerly-upwind position, the seventh-ranked station, Mont  limar at 0.695, being the one dropped. Ranking the same stations by their correlation with the *change* target gives a different order, in which Mont  limar (0.110) and Perpignan (0.125) outscore M  nchen (  0.071) and Kassel (0.067). The set is a physical choice, not the one the change target would have made on its own.

- **controlled, off by default:** day-of-year sin / cos (2) and missing-value indicators (163). These appear only as ablation arms, so that no calendar information enters the main configuration.

The main input is levels + deltas, $p = 326$; adding grads gives 333. Two reduced inputs serve the ablations: the upstream-7 subset (14 columns, upstream $T_{\max}$ levels and deltas) and the own-station pair (Basel $T_{\max}$ level and delta, $p = 2$).

4.3 Model ladder

Baselines. *Zero-change* predicts $\hat{y} = 0$ and sets the admission bar of Eq. 2. *Yesterday-  * repeats today’s change; it should fail, and its failure documents that the change series has no exploitable autocorrelation. *Monthly-   climatology* uses the training-period mean change per calendar month and should land near zero skill.

Ridge, in closed form. For each feature set we solve

$$\hat{w} = \arg \min_w \|Xw - y\|_2^2 + \alpha \|w\|_2^2 = (X^\top X + \alpha I)^{-1} X^\top y \quad (3)$$

exactly, by `torch.linalg.solve`. There is no learning rate, no epoch count and no random initialisation, so the answer is exact to machine precision and identical on every run. A single hyperparameter α is selected on a 25-point logarithmic grid from 10^{-2} to 10^4 by validation MAE on 2008. The inputs and the target are mean-centred and the intercept is left unpenalised, so the penalty acts on w alone. Eq. 3 drops the $1/n$ of the mean-squared risk; that only rescales α relative to Eq. 1’s convention, and the grid is searched rather than fixed. With 2920 training samples and $p \leq 333$ the solve is instantaneous, which makes a programme of several hundred regularisation fits cheap and exactly repeatable.

Lasso. The L_1 counterpart, $\arg \min_w \|Xw - y\|_2^2 + \alpha \|w\|_1$, has no closed form and is trained with Adam over 200 epochs, α selected on 2008 from 8 geometric values in $[10^{-3}, 1]$. It uses the same mean-squared-error convention as Eq. 3 and serves as a second regularisation family. As a selection device it does not work: at the validation-optimal α , only 2 of 326 coefficients fall below the numerical threshold of 10^{-4} , so the fit is a differently regularised linear model rather than a sparse one.

MLP. A small feed-forward network over hidden sizes $\{32\}$, $\{64, 32\}$ or $\{128, 64\}$, with GELU activations, dropout $\in \{0.2, 0.4\}$ and weight decay $\in \{10^{-4}, 10^{-3}\}$, twelve configurations in all. Training uses the Huber loss with $\delta = 1$ on targets standardised by the training-period σ_Δ , Adam at lr 10^{-3} and batch 64, and early stopping on validation MAE with patience 30 for at most 300 epochs. Each configuration runs three fixed seeds $\{0, 1, 2\}$, selection is by mean validation MAE, and the reported test figure is the mean of the three seeds’ predictions.

Multi-station extension. The 18 stations’ change series are stacked into 65,736 station-days, each carrying the same global 326-column field and its own station’s $\Delta T_{\max}$ as target. Change space removes station climatologies, which is what makes pooling physically admissible here in a way it would not be at level space.

Everything is PyTorch 2.13 with CUDA, and seeds are fixed throughout: 42 for the bootstrap, 0 for the Lasso gradient fit, $\{0, 1, 2\}$ for the MLP. One mismatch is worth naming. MAE is the scoring metric, and its optimal predictor is the conditional *median*, whereas the linear models estimate a conditional *mean* by least squares and the MLP’s Huber loss reduces to a mean in its central region. The penalties are also chosen on validation MAE while the fits minimise squared error. This does not invalidate the comparison, since all models are scored the same way, but it does mean each model is not fitted to the loss it is judged by.

5 Experiments

5.1 Main results

Table 2: Test-period results for $\Delta T_{\max}(t+1)$ at Basel, 2009-01-01 to 2010-01-01 (366 days). MAE in $^{\circ}\text{C}$ with 95% moving-block bootstrap CIs (7-day blocks, 1000 resamples). Skill is defined in Eq. 2; corr is the Pearson correlation between predicted and observed changes.

Model	MAE [95% CI] ($^{\circ}\text{C}$)	RMSE ($^{\circ}\text{C}$)	R^2	Corr	Skill vs. zero
Zero-change (bar)	2.393 [2.104, 2.680]	3.176	0.000	—	—
Yesterday- Δ	3.447 [3.033, 3.916]	4.647	-1.141	-0.072	-44.0%
Monthly- Δ climatology	2.393 [2.098, 2.682]	3.174	0.001	0.034	+0.0%
Ridge: levels only (163)	1.845 [1.651, 2.033]	2.383	0.437	0.665	+22.9%
Ridge: levels+ Δ (326)	1.827 [1.623, 2.014]	2.400	0.429	0.660	+23.7%
Ridge: levels+ Δ +grad (333)	1.829 [1.624, 2.012]	2.410	0.424	0.660	+23.6%
Ridge: upstream-7 subset (14)	2.056 [1.829, 2.284]	2.760	0.245	0.502	+14.1%
Ridge: own station only (2)	2.392 [2.104, 2.675]	3.152	0.015	0.183	+0.0%
Lasso: levels+ Δ (326)	1.773 [1.574, 1.955]	2.365	0.445	0.671	+25.9%
MLP ensemble (best config)	1.751 [1.546, 1.946]	2.341	0.457	0.682	+26.9%

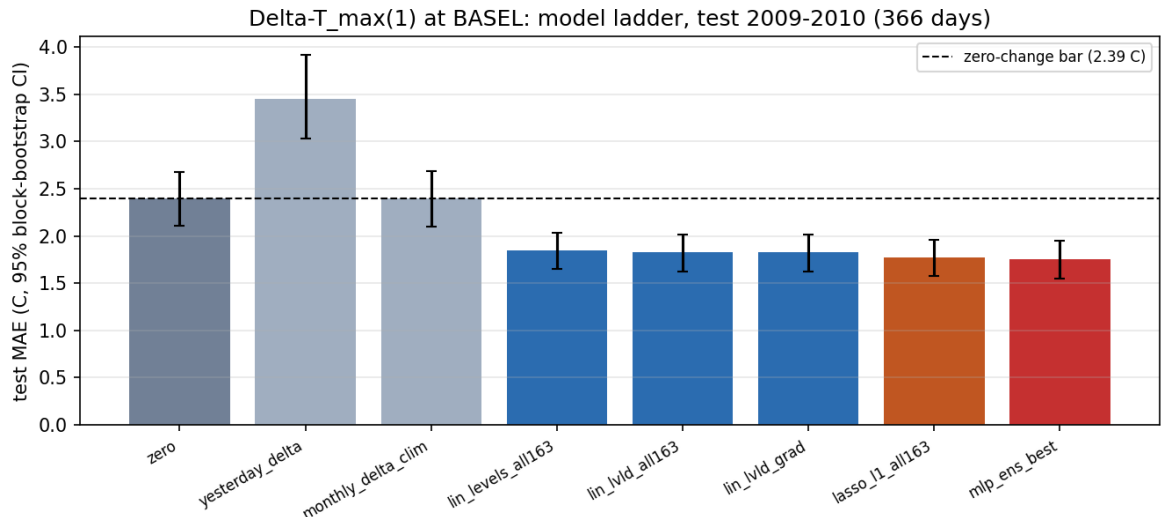


Figure 1: Model ladder on the test window. Bars: test MAE in $^{\circ}\text{C}$; whiskers: 95% block-bootstrap CIs. Grey: naive baselines; blue: ridge variants; orange: Lasso; red: MLP ensemble. The dashed line marks the zero-change bar at 2.393 $^{\circ}\text{C}$.

All three learned models beat the zero-change baseline with paired-bootstrap differences comfortably below zero (Table 3), so the skill is not a fluctuation. The differences among the

learned models are not significant: the MLP improves on ridge by 0.076°C with a 95% interval of $[-0.149, 0.000]$. Information matters more than model class. The full field beats the upstream-7 subset by a significant 0.229°C , while the target station’s own temperature observations give $+0.0\%$ skill.

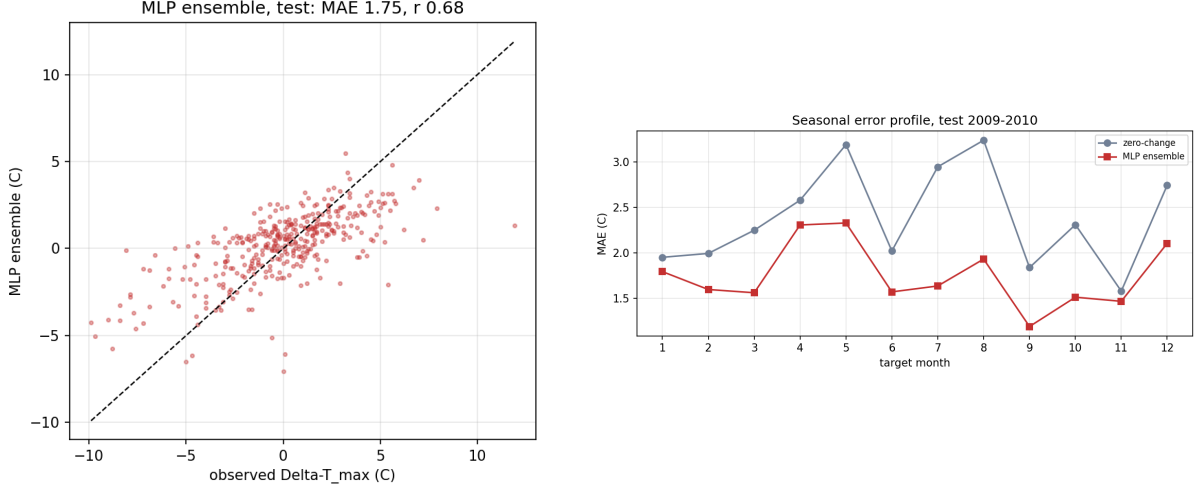


Figure 2: Left: MLP-ensemble predictions against observed $\Delta T_{\max}$ on the test window (MAE 1.75°C , $r = 0.68$); moderate changes are captured, extremes are shrunk. Right: monthly error profile; both the zero-change baseline (grey) and the MLP (red) err most in late spring and summer, where the mean absolute change is largest, and least in autumn and winter.

The shrinkage in Fig. 2 (left) is not an MLP defect. A model that cannot resolve the drivers of individual frontal passages minimises expected loss precisely by shrinking. What the shrinkage does is cap the achievable MAE, and it points to the seasonal structure: the skill gap between model and baseline is widest in the months of largest mean absolute change, which on this particular test year falls in late spring and summer (right).

5.2 Pairwise significance

Table 3: Paired block-bootstrap differences in test MAE ($A - B$, in $^{\circ}\text{C}$; negative means A is better). “sig” marks intervals excluding zero.

A – B	diff ($^{\circ}\text{C}$)	95% CI	sig
MLP – zero	−0.642	[−0.843, −0.467]	✓
Lasso – zero	−0.620	[−0.834, −0.416]	✓
Ridge (lvl+ Δ) – zero	−0.566	[−0.769, −0.383]	✓
MLP – ridge (lvl+ Δ)	−0.076	[−0.149, 0.000]	
MLP – Lasso	−0.023	[−0.119, 0.068]	
Lasso – ridge (lvl+ Δ)	−0.054	[−0.101, 0.002]	
Ridge (lvl+ Δ) – ridge (up7)	−0.229	[−0.367, −0.115]	✓

The MLP-versus-ridge interval touches zero at its upper bound, so the MLP is consistent with no improvement beyond linear. At 366 test days, effects below roughly 0.1°C cannot be resolved.

5.3 Feature ablations

Levels and deltas each carry most of the signal alone and add little to each other, 22.9% and 17.4% individually against 23.7% combined, which says the two blocks largely encode the same

Table 4: Feature-set ablation under the same protocol; ridge except where noted. Dims is the number of input columns.

Feature set	Dims	MAE (°C)	Skill
levels only	163	1.845	+22.9%
deltas only	163	1.978	+17.4%
levels + deltas (main)	326	1.827	+23.7%
+ pressure grads	333	1.829	+23.6%
+ grads + day-of-year sin / cos	335	1.834	+23.4%
+ grads + missing indicators	496	1.842	+23.0%
upstream-7: levels + deltas	14	2.056	+14.1%
upstream-7: deltas only	7	2.115	+11.6%
upstream-7: deltas + grads	14	2.075	+13.3%
own station: level + delta	2	2.392	+0.0%
Lasso, full (326), val- α	326	1.773	+25.9%
Lasso top-25 features, ridge refit	25	1.859	+22.3%

weather systems. Adding day-of-year features lowers skill to 23.4%, which is the evidence that the change target holds no calendar free lunch. Pressure gradients are redundant once the full field is present, 23.6% against 23.7%, but add value inside the upstream subset, 11.6% rising to 13.3%, so they carry partly overlapping information.

The missing-indicator arm is nearly uninformative. At most 118 of 595,602 cells could ever be flagged, 0.02%, so an arm of that size has almost no statistical power. Three of the fifteen come from columns we masked: the Tours pressure, at one cell in 2008, so it reaches the validation predictions only, and the Stockholm pressure, at three cells all inside the training window. Their change columns inherit two and six imputed entries respectively, under 0.02% of rows, filled with training-period medians. The single test-period defect is removed before differencing.

Sparsity yields a compact core. A ridge refit on the Lasso’s top-25 features retains 22.3 of the full model’s 23.7 percentage points of skill, so a 25-feature model captures nearly all of it. But the Lasso itself keeps 324 of 326 coefficients non-zero at the validation-optimal α , so the signal is distributed across the field rather than concentrated in a handful of channels.

5.4 Where the weight mass sits

Figure 3 shows what the fit leans on. Ordered by standardised magnitude, the largest of the fifteen are the day-to-day changes of pressure at Heathrow (-0.30), Basel ($+0.28$) and Mont  limar ($+0.23$), the pressure level at Tours (-0.17), the change in humidity at Maastricht ($+0.13$), the pressure level at Heathrow (-0.12) and at Basel ($+0.10$).

Two things limit what the figure can carry. The fifteen are selected by raw coefficient magnitude, and pressure has a raw standard deviation of only 0.008, so it climbs that ranking partly on its units. The claim that survives concerns the weight mass as a whole rather than any single channel. With 326 standardised and near-collinear inputs the minimiser is unique for every $\alpha > 0$, yet the data cannot separate near-collinear alternatives, so no individual coefficient is stably interpretable.

5.5 MLP grid

The grid is flat. Every configuration’s test ensemble MAE falls between 1.70 and 1.75°C across a parameter range from about 10.5k to 50k. The validation-selected configuration, 64–32 hidden units with dropout 0.4 and weight decay 10^{-4} and mean validation MAE 1.797°C, gives per-seed test MAEs of 1.795, 1.758 and 1.757°C (seed standard deviation 0.018) and 1.751°C for the three-seed ensemble. Capacity is not the binding constraint at this sample size.

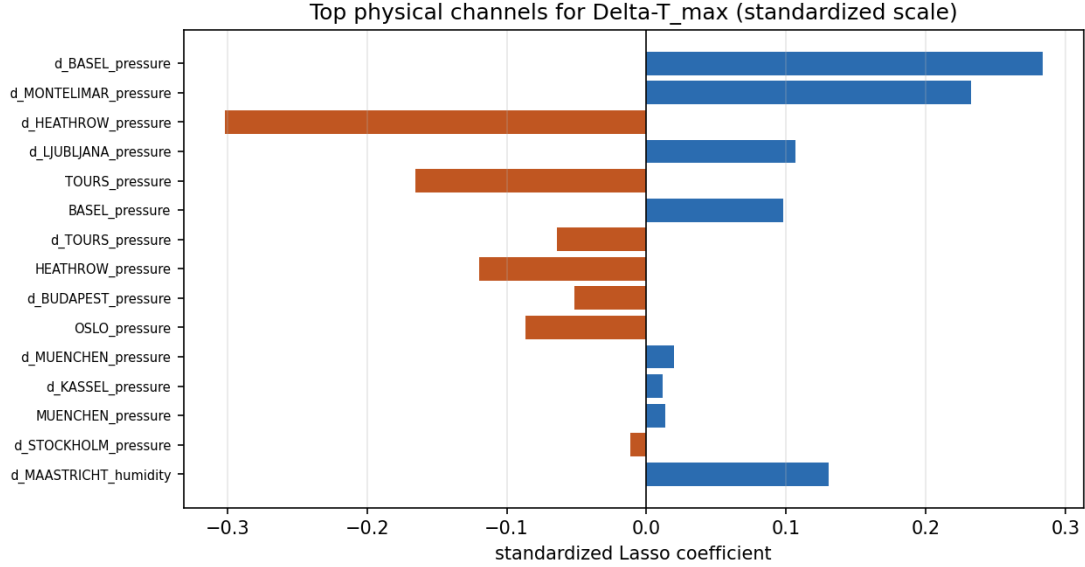


Figure 3: Fifteen Lasso coefficients with the largest absolute value in raw units, plotted on the standardised scale. Blue: positive, orange: negative. Fourteen of the fifteen are pressure features, at Basel and elsewhere; the exception is the change in humidity at Maastricht.

5.6 Multi-station extension

Table 5: Multi-station extension, 6588 test station-days in 2009. Per-station models are 18 independent ridge fits on the full field; pooled models are single global fits with per-station targets and no station identity among the inputs. Skills in this table are measured against the pooled zero-change bar; Figure 4 uses each station’s own bar, so the two are not directly comparable.

Model	MAE ($^{\circ}\text{C}$)	95% CI	Skill vs. pooled zero
Pooled zero-change	2.187	—	—
Per-station ridge (18 models, mean)	1.705	[1.299, 1.998] [†]	+22.0%
Pooled ridge (one global model)	2.066	[2.016, 2.115]	+5.5%
Pooled MLP	2.038	[1.986, 2.088]	+6.8%

[†]range across stations, not a bootstrap CI.

Pooling costs accuracy. One global ridge reaches 2.066°C against 1.705°C for the 18 independent models, a gap of 0.36°C that exceeds the pooled model’s own interval and stands above the best single station. A pooled MLP closes part of the gap to 2.038°C .

The mechanism is station specificity. Each station’s change series responds to the same synoptic field through its own aspect, elevation and continentality, and one weight vector cannot express 18 different response vectors. The qualification matters: with no station identifier among the inputs, this failure is close to inevitable for a linear model. Exact restoration of the independent models would need a separate slope vector per station, that is 18 sets of station-by-feature interaction columns rather than one-hot station columns, which would give station-specific intercepts only. What the experiment rules out is unconditioned parameter sharing, not shared representations in general. That the pooled MLP recovers part of the gap suggests the station information is recoverable without an explicit identifier, though this report does not localise it. The control that would settle the question, a shared trunk with station-specific heads, was not run.

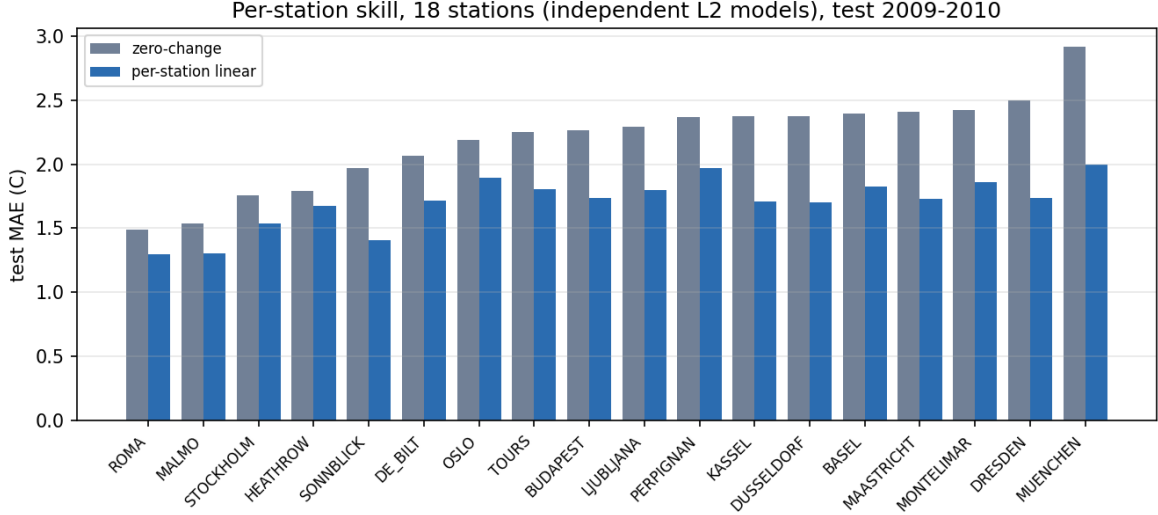


Figure 4: Per-station test MAE, 366 days each: zero-change baseline (grey) against an independent ridge per station (blue), sorted by baseline difficulty. Skill is positive at all 18 stations, ranging from +6.7% at Heathrow to +31.4% at München.

6 Discussion and Limitations

The three learned models cluster at 24–27% skill with statistically indistinguishable differences, and we read that plateau as a property of the task rather than of the models. Once persistence and the calendar are removed, most of what remains is synoptic noise at the day-ahead horizon, and roughly a quarter of MAE reduction is what the available information supports. The two baselines that fail say where the signal is not: it lies in the spatial field, not in the target’s own history. The MLP’s advantage over the exact linear solution sits inside the same noise. Nothing in the validation curve points deeper, since across the twelve configurations of Section 5.5 validation MAE does not fall with capacity and test MAE spans 0.05°C. The selected configuration is in fact the weakest of the twelve on test, which is what selecting on validation buys when the differences are this small. Within that range the network and the linear solution are the same predictor up to noise, and 366 test days cannot separate them.

The upstream processing deserves a straight accounting. Mean imputation of sparse columns compresses variance and inflates correlations in exactly those columns, and our own median imputation of the masked cells is a second such manipulation, acting on 0.02% of cells. The upstream imputation is a different matter: it touched up to 5% of the cells in every column it filled, an unrecorded total, and it is the reason the defect audit in Section 2 had to be run from scratch.

Limits. A 366-day window bounds what can be claimed, and effects below roughly 0.1°C are out of reach. The multi-station stack of 6588 station-days has more power but answers a different question, that of per-station response heterogeneity. About thirty configurations were scored on this one window with no correction for multiple comparisons, so which model family leads this account is a post-hoc choice. All preprocessing statistics are training-only, but feature construction is weaker: the one-day difference window was chosen after seeing the horizon RMSE on the 2009–2010 targets, which is the one place test information entered the design. The station ranking used training-window correlations only, and no model was fitted on test data during selection, but a fully held-out region would be cleaner. Beyond these, the study covers one target station and one test year: the per-station analysis of Fig. 4 speaks to generality across stations, not across years, and the $T_{\max}$ series is a homogenised product whose upstream quality control

we take as given.

On the advection reading. The upstream advantage and the concentration of weight mass on pressure are both consistent with mid-latitude advection, and both are equally consistent with a direction-free spatial signal. No test here separates the two. The pressure-gradient features, which are the most direct proxy for along-flow structure, add nothing once the full field is present, which if anything counts against the advection reading. A geostrophic-wind computation compared against a domain-mean-pressure baseline would be the way to settle it. Multi-horizon skill decay, $h = 1 \dots 10$, would map the predictability limit quantitatively, next-day precipitation occurrence is the natural low-skill classification extension under the same protocol, and the pooled result points directly at replacing the single global model with station-conditional heads.

7 Conclusion

Predicting the day-to-day change rather than the level removes persistence and the seasonal cycle from the reward function and makes the zero-change forecast an explicit bar. Under the chronological split, with training-only statistics and block-bootstrap uncertainty, a ladder of deliberately simple models recovers 24–27% MAE skill on the test year. Ridge reaches 1.827°C, Lasso 1.773°C and the MLP ensemble 1.751°C, against 2.393°C for zero-change. Every improvement over the baseline is significant; every difference among the learned models is inside the noise. The signal is spatial and weighted towards pressure. The target station’s own temperature carries none of it, while its own pressure is among the strongest single predictors.

What makes the result usable rather than merely high-scoring is that each of those claims is a falsifiable number with an interval attached, and several of them point the other way: the null result on the MLP gain, the redundant gradient features, the pooling result that only holds without station conditioning.

Data availability

The dataset is public under the MIT licence: https://github.com/florian-huber/weather_prediction_dataset, release commit 83d70ee, archived on Zenodo (doi:10.5281/zenodo.7525955). Code and machine-readable results accompany this report; Appendix A lists the scripts that regenerate every table and figure.

References

- [1] Klein Tank, A.M.G., et al. (2002). Daily dataset of 20th-century surface air temperature and precipitation series for the European Climate Assessment. *International Journal of Climatology*, 22, 1441–1453.
- [2] Huber, F., van Kuppevelt, E.D., Steinbach, P., Sauzé, C., Liu, Y., Weel, B. (2023). Will the sun shine? — An accessible dataset for teaching machine learning and deep learning. *Proceedings of the Third Teaching Machine Learning and Artificial Intelligence Workshop*, PMLR 207, 27–31.

A Reproducibility

Environment. Python 3.12, PyTorch 2.13 with CUDA, NumPy and pandas; scikit-learn appears only in the horizon and station probes. No randomised splitting anywhere, and seeds fixed as described in Section 4.3.

Regenerating the results. With the dataset CSV placed under `data/`, from the repository root:

<code>analysis/eda.py</code>	dataset audit behind Section 2
<code>analysis/formulation_probe.py</code> (Table 1, Section 4.2)	horizon decay and upstream-station correlations
<code>analysis/delta_t_probe.py</code>	sensitivity of the change target
<code>analysis/r1_baselines.py</code>	pipeline and baseline table
<code>analysis/r2_lasso_ablation.py</code>	Table 4 and the coefficients behind Figure 3
<code>analysis/r3_mlp.py</code>	MLP grid (Section 5.5) and Table 5
<code>analysis/r4_eval_figures.py</code>	Tables 2–3 and Figures 1–4

Each script writes its tables as CSV, and every number in this report traces to one of those files.

Attribution. Observations derive from ECA&D station data [1]; dataset and context from Huber et al. [2]. Upstream processing is described in Section 2: columns with more than 5% missing were dropped, the rest mean-imputed, and units scaled without standardising.